

JOHN PATRICK COLLINS, M.S.

BIOINFORMATICS DATA SCIENTIST | SCIENTIFIC SOFTWARE ENGINEER

PROFILE

- ✳ **Bioinformatics scientist and scientific software engineer with 10+ years of experience** translating novel assays and complex biological data into validated, reproducible workflows and scientist-facing products across clinical diagnostics, immuno-oncology, single-cell genomics, and long-read synthetic biology.
- ✳ **Advanced Python developer** with end-to-end ownership from requirements and data modeling through implementation, benchmarking, QC, documentation, and operational handoff.
- ✳ **Hands-on with NGS/ONT, Benchling/LIMS, cloud/HPC, and supervised AI-assisted engineering.**

EXPERIENCE

INDEPENDENT COMPUTATIONAL BIOLOGY Researcher / Open-Source Developer

2025 - PRESENT

- Lead ongoing [CORD research](#) across 43 public perturbational single-cell datasets, evaluating recurrent response coordinates against external DepMap CRISPR GeneEffect data and TCGA tumor projections using explicit study- and context-holdout validation.
- Architect reusable Python/Colab workflows with versioned inputs, content-hash caches, deterministic scoring, run/release manifests, claim-evidence ledgers, and leakage-aware holdout evaluation.
- Use AI coding and reasoning systems to accelerate implementation, testing, review, failure analysis, and documentation while retaining human control of architecture, biological interpretation, provenance, security, and scientific claims.

BACSTITCH DNA, INC.

Staff Software Developer

MARCH 2024 - NOVEMBER 2024

- Co-designed and implemented Nanoplex, the Python/ONT pipeline used to verify [SCRIVENER](#) assemblies: combinatorial-barcode demultiplexing, minimap2/samtools alignment, bcftools/Sniffles2 variant calling, SciPy read-length clustering, and automated QC/purity scoring on a 36-core cluster.¹
- Analyzed sequence-verification data for a Cell Systems study of >5,000 DNA assemblies (2-19 blocks; constructs to 81 kb), generated publication evidence, edited the manuscript, and coauthored the paper.
- Built Benchling API tools and data models for assembly planning, sample lineage, inventory/entity creation, and LIMS registration; partnered across R&D and software on infrastructure and CI planning.

CONTACT

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GITHUB: github.com/jcollins-bioinfo

EDUCATION

2013 - 2014 **BIOMOLECULAR ENGINEERING AND BIOINFORMATICS (M.S.),**
UNIVERSITY OF CALIFORNIA, SANTA CRUZ

2009 - 2013 **BIOENGINEERING (B.S.),**
Graduated With Honors in the Major,
UNIVERSITY OF CALIFORNIA, SANTA CRUZ

CORE EXPERTISE

SCIENTIFIC SOFTWARE. Python (NumPy, SciPy, pandas, Polars, scikit-learn), R, Bash/Linux, Git; pipelines, APIs, ETL/data models, and testing.

COMPUTATIONAL GENOMICS. NGS/ONT, RNA-seq/scRNA-seq, cfDNA, V(D)J; demultiplexing, alignment, assembly, variant calling, and QC.

VALIDATION & ANALYSIS. Statistical modeling, clustering, permutation tests, benchmarking, leakage-aware evaluation, and reproducibility.

PLATFORMS & APPLICATIONS. Benchling LIMS/ELN/API; AWS, Azure, GCP; Docker; Linux/HPC; Dash/Plotly and Flask.

AI-ASSISTED ENGINEERING. Specifications, task decomposition, code review, tests/evals, provenance, and human verification.

PEER-REVIEWED PUBLICATIONS

[1] Matsui T, Hung P-H, Mei H, Liu X, Li F, Collins J, et al. **High-throughput DNA engineering by mating bacteria.** Cell Systems. 2026:101656. [doi:10.1016/j.cels.2026.101656](https://doi.org/10.1016/j.cels.2026.101656)

[2] Grskovic M, Hiller DJ, Eubank LA, Sninsky JJ, Christopherson C, Collins JP, et al. **Validation of a Clinical-Grade Assay to Measure Donor-Derived Cell-Free DNA in Solid Organ Transplant Recipients.** J Mol Diagn. 2016;18(6):890-902. [doi:10.1016/j.jmoldx.2016.07.003](https://doi.org/10.1016/j.jmoldx.2016.07.003)

EXPERIENCE (CONTINUED)

CALICO LIFE SCIENCES

Data Management Engineer, Contract

SEPTEMBER 2023 - NOVEMBER 2023

- Built Python ETL, normalization, and validation automation for migrating laboratory inventory into Benchling, preserving consistent entity representations and enabling reliable downstream lab-management workflows.

PACT PHARMA, INC.

Bioinformatics Data Scientist

SEPTEMBER 2018 - SEPTEMBER 2020

- Developed Python-centric NGS and Sanger analysis software for personalized adoptive T-cell immunotherapy and a patient-specific phase I clinical program.³
- Designed Benchling chain-of-identity and automated data-capture workflows linking patient, sample, construct, sequencing, and analysis records.
- Built a Dash application for gene-edited plasmid sequencing that reduced routine processing from hours to minutes and delivered interactive QC and results to scientists.

BRISTOL MYERS SQUIBB

Bioinformatics Pipeline Developer, Contract

FEBRUARY 2018 - JUNE 2018

- Integrated open-source bioinformatics tools with custom Python into a parallelized V(D)J gene-rearrangement analysis application for an HPC environment.

BIO-RAD LABORATORIES

Bioinformatician II - Sequencing Analysis

APRIL 2017 - JULY 2017

- Developed secondary and tertiary NGS analysis for a single-cell whole-transcriptome RNA-seq assay.
- Automated cloud-based analytical reports using the Python scientific stack.

CAREDX, INC.

Bioinformatics Analyst

NOVEMBER 2014 - APRIL 2017

- Engineered Python R&D software and operated NGS pipelines for donor-derived cell-free DNA, supporting analytical validation and assay-performance development for AlloSure, a noninvasive solid-organ transplant diagnostic.^{2,4,5}
- Performed MiSeq library preparation for PCR-based dd-cfDNA quantification, connected wet-lab observations to pipeline QC, statistical analysis, and interpretation, and coauthored the clinical-grade assay validation paper (including creation of all figures).

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Undergraduate & Graduate Student Researcher

2011 - 2014

- Contributed to single-cell RNA-seq, nanopore sequencing, and chromosome-protein dynamics research in the Pourmand, Akesson, and Bhalla laboratories, respectively.

METHODS & TOOLS

GENOMICS. minimap2, BWA, samtools, bcftools, Sniffles2, GATK, Dorado, Clustal Omega; UCSC Genome Browser, Ensembl, and NCBI.

DATA & ML. NumPy, SciPy, pandas, Polars, scikit-learn; statistical modeling, clustering, permutation tests, representation learning, experimental design, and scientific visualization.

DELIVERY. Requirements, implementation, benchmarking, automated QC, documentation, cross-functional review, and operational handoff.

AI & REPRODUCIBILITY

- Apply frontier AI coding and reasoning systems across scientific-software development: write explicit specifications, decompose tasks, review generated code, design tests/evaluations, diagnose failures, and document decisions; retain human ownership of architecture, biological interpretation, falsification, security boundaries, provenance, and scientific claims.
- Build reproducibility into delivery through versioned inputs, deterministic execution, leakage-aware validation, traceable claims, and reviewable release artifacts.

PUBLISHED CONFERENCE ABSTRACTS

[3] Dalmas O, Pan Z, Shieh C, et al., including Collins J. **A high-throughput platform to produce neoE-HLA libraries for capturing neoE-specific T cells from the peripheral blood of patients with solid tumors.** Cancer Res. 2020;80(16 Suppl):Abstract 3253. [doi:10.1158/1538-7445.AM2020-3253](https://doi.org/10.1158/1538-7445.AM2020-3253)

[4] Crespo-Leiro M, Hiller D, Woodward R, Grskovic M, Marchis C, Song M, Collins J, Zuckermann A. **Analysis of Donor-Derived Cell-Free DNA with 3-Year Outcomes in Heart Transplant Recipients.** J Heart Lung Transplant. 2017;36(4):S69-S70. [doi:10.1016/j.healun.2017.01.172](https://doi.org/10.1016/j.healun.2017.01.172)

[5] Khush KK, Grskovic M, Luikart H, et al., including Collins J. **Circulating Cell-Free DNA as a Non-Invasive Marker of Pediatric Heart Transplant Rejection and Immunosuppressive Treatment.** J Heart Lung Transplant. 2016;35(4):S75. [doi:10.1016/j.healun.2016.01.205](https://doi.org/10.1016/j.healun.2016.01.205)